

Project Brief

Project Title

Automation of Wholesale Coffee Orders for Volta Coffee

Project Overview

This project develops an AI-powered workflow using n8n to automate the processing of wholesale coffee orders for Volta Coffee Roasters. Customer orders are submitted as unstructured free-text messages and are automatically analyzed, classified, and routed to the appropriate processing path. The workflow combines artificial intelligence with rule-based automation to reduce manual effort while ensuring that business-critical decisions remain under human supervision.

Business Problem

Volta Coffee currently processes wholesale orders manually. Every incoming order must be read, interpreted, and evaluated before deciding how it should be handled. This process is time-consuming, repetitive, difficult to scale and dependent on a single employee's experience. Furthermore, manually processed orders do not create structured data for reporting or future automation.

Project Objectives

- Receive wholesale orders through a Webhook.
- Analyze free-text orders using Google Gemini.
- Extract structured customer and product information.
- Classify each order automatically.
- Store standard orders in Airtable.
- Create Gmail confirmation drafts.
- Notify the sales team through Slack when manual review is required.
- Support a Human-in-the-Loop workflow.

Solution Overview

1. Receive incoming order via Webhook.
2. Standardize the incoming data.
3. Analyze the order using an AI Agent with Google Gemini.
4. Validate the output using a Structured Output Parser.
5. Classify the order as standard, needs_review or needs_clarification.
6. Execute the corresponding business logic (Airtable, Gmail or Slack).

Technologies Used

n8n, Google Gemini Chat Model, Structured Output Parser, Airtable, Gmail, Slack, JSON, Webhooks.

Expected Benefits

Reduced manual effort, standardized order processing, structured data storage, faster handling of routine orders, Human-in-the-Loop support and improved scalability.

Project Outcome

The project demonstrates how AI and workflow automation can efficiently process wholesale orders while keeping critical decisions under human supervision.